

Department of Computer Science and Engineering(CSE)

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

Course code:CSE-3524

Course Title: Microprocessor & Microcontroller Sessional

EXPERIMENT 4

Experiment Name: To observe the operation of logic instructions.

OBJECTIVES:

1. To realize the operation of logic instructions like AND, OR & XOR.
2. To realize the operation of logic instructions like SHIFT and ROTATE.
3. To check the register values in kit box.
4. To analyze the result.

NECESSARY EQUIPMENT:

1. Microsoft MACRO assembler (MASM) to run the program.
2. Notepad to write the program.
3. 8086 microprocessor kit.

PROBLEM STATEMENT:

1. Perform logic operation AND, OR, XOR instruction on the content of register AX, BX and realize their operation.

Program1: Logic instruction 1

```
.MODEL SMALL
.CODE

MOV AX, 55BBH
MOV BX, 5566H
AND AL, 5BH
OR AH, 09H
XOR BX, 333FH
XOR BX, BX
END
```

DATA TABLE:-

Instruction	AX	BX	CX	DX	Set Flag
MOV AX, 55BBH					
MOV BX, 5566H					
AND AL, 5BH					
OR AH, 09H					
XOR BX, 333FH					
XOR BX, BX					

REPORT WRITING:-

Include the experiment name, objectives, necessary software/ equipment and problem statement.

- 1. Write the program with brief explanation.
- 2. Insert the printed copy of “list file” of the program.
- 3. Find the result of all related registers from kit box for each line of the program.
- 4. Verification of logical operation by theoretical calculation.
- 5. Finding the active flags from flag register and comparing them with the calculated values.
- 6. Draw all the shifting and rotating steps and verify the result and also the active flags.
- 7. Write a discussion including all the key points from the result of the program.
- 8. Write a discussion if you have any error in the program or troubleshooting of the program.
- 9. Write the answer of the related questions given below.

RELATED QUESTIONS:-

- 1. What is the main application of AND, OR and XOR instruction?
- 2. If the count value in program2 is not given by CL register, then what problem does occur?
- 3. What are the basic differences between arithmetic and logical instructions?

EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.exe_ -- emu8086 assembler version: 4.08

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=====			
=====			
[LINE]	LOC:	MACHINE CODE	SOURCE
=====			
=====			
[1]	:		.MODEL SMALL
[2]	:		.CODE
[3]	0000:	B8 BB 55	MOV AX, 55BBH
[4]	0003:	BB 66 55	MOV BX, 5566H
[5]	0006:	24 5B	AND AL, 5BH
[6]	0008:	80 CC 09	OR AH, 09H
[7]	000B:	81 F3 3F 33	XOR BX, 333FH
[8]	000F:	33 DB	XOR BX, BX
[9]	:		END
[10]	:		